

Normal Distribution



Properties of the normal distribution

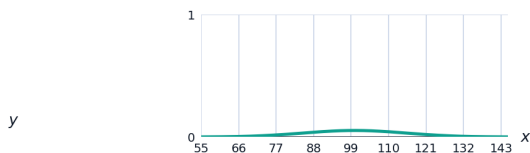
- 1 State two key properties of the normal distribution curve.
- 2 Explain what it means for the normal distribution curve to be "asymptotic" to the horizontal axis.
- 3 State one similarity and one difference between the normal distribution and the discrete uniform distribution.

Standardizing a normal variable

- 4 A random variable $X \sim N(50, 64)$. Find the z-score corresponding to $x = 62$, using $z = \frac{x - \mu}{\sigma}$.
- 5 A random variable $X \sim N(30, 25)$. Find the z-score corresponding to $x = 23$.
- 6 A random variable $X \sim N(70, 144)$. Find the z-score corresponding to $x = 88$.

Finding probabilities using the standard normal table

- 7 A random variable $X \sim N(100, 225)$. Find $P(X < 115)$, given that $\Phi(1) \approx 0.8413$.



- 8 Using the same distribution $X \sim N(100, 225)$, find $P(X > 130)$, given that $\Phi(2) \approx 0.9772$.
- 9 A random variable $X \sim N(50, 16)$. Find $P(X < 54)$, given that $\Phi(1) \approx 0.8413$.
- 10 A random variable $X \sim N(80, 100)$. Find $P(X > 95)$, given that $\Phi(1.5) \approx 0.9332$.

Finding a value given a probability (inverse normal)

- 11 A random variable $X \sim N(60, 100)$. Find the value x such that $P(X < x) = 0.9332$, given that $\Phi(1.5) \approx 0.9332$.

- 12 A random variable $X \sim N(80, 64)$. Find the value x such that $P(X < x) = 0.9772$, given that $\Phi(2) \approx 0.9772$.
- 13 A random variable $X \sim N(45, 49)$. Find the value x such that $P(X < x) = 0.8413$, given that $\Phi(1) \approx 0.8413$.
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The probability of a value lying between two bounds

- 14 A random variable $X \sim N(40, 25)$. Find $P(35 < X < 48)$, given that $\Phi(1) \approx 0.8413$ and $\Phi(1.6) \approx 0.9452$.
- 15 A random variable $X \sim N(20, 16)$. Find $P(16 < X < 26)$, given that $\Phi(1) \approx 0.8413$ and $\Phi(1.5) \approx 0.9332$.
- 16 A random variable $X \sim N(50, 100)$. Find $P(35 < X < 65)$, given that $\Phi(1.5) \approx 0.9332$.
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The normal approximation to the binomial distribution

- 17 A random variable $X \sim B(200, 0.4)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid (both np and $n(1 - p)$ should exceed 5).
- 18 A random variable $X \sim B(150, 0.5)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid.
- 19 A random variable $X \sim B(100, 0.3)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid.
- 20 A random variable $X \sim B(250, 0.2)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid.
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Solving for unknown parameters of a normal distribution

- 21 A random variable $X \sim N(\mu, 36)$ satisfies $P(X < 50) = 0.8413$, given $\Phi(1) \approx 0.8413$. Find μ .
- 22 A random variable $X \sim N(\mu, 49)$ satisfies $P(X < 66) = 0.9772$, given $\Phi(2) \approx 0.9772$. Find μ .
- 23 A random variable $X \sim N(\mu, 25)$ satisfies $P(X < 40) = 0.9332$, given $\Phi(1.5) \approx 0.9332$. Find μ .
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A well-designed word problem: setting a pass mark for an exam

- 24 This question has two parts. Scores on a standardized exam are normally distributed with mean $\mu = 68$ and standard deviation $\sigma = 12$. The exam board wants to set a pass mark so that exactly the top 10% of students pass, and is told that the z-value corresponding to the top 10% (i.e. $P(Z > z) = 0.10$) is $z \approx 1.2816$. (a) Find the pass mark, to the nearest whole number. (b) A student scores 90 on the exam. Find their z-score, and state whether they are in the top 10%.